C6.5 Interactions with main memory

This section describes the circumstances in which the interconnect:

- must read or update main memory directly
- can pass permission to update main memory to a master.

描述了Interconnect需要读取或者更新main memory的情况

It contains to following sections:

- *Interconnect read from main memory or peripheral device*
- *Main memory update generated by the interconnect* on page C6-235
- *Permission to update main memory* on page C6-236.

C6.5.1 Interconnect read from main memory or peripheral device

The interconnect must always read from main memory, or the appropriate peripheral device, for a ReadNoSnoop transaction.

对于ReadNoSnoop请求，Interconnect都需要从main memory或者外设设备中读取数据（对于多层Cache，可以是L2）

If the interconnect has not obtained the required data from a snoop transaction, the interconnect must read from main memory to complete the following transactions:

Interconnnct在下面这几中请求中，如果发现没有任何一个Cache设备提供了数据，那么就需要从main memory来读取数据了

- ReadOnce
- ReadClean
- ReadNotSharedDirty
- ReadShared
- ReadUnique.

Interconnect允许在所有的Snoop请求完成之前发起读取main memory的请求

An interconnect is permitted to read from main memory before all snoop transactions have completed. However, the following rules apply:

- Data obtained from main memory must not be used if any cache in the shareability domain of the master is holding a dirty copy of the cache line. Therefore, if the cache line is provided by a snoop transaction, then data obtained from main memory must not be used.

  如果通过Snoop返回的CacheLine是Dirty的，那么通过读取main memory得到的数据就不能被使用（因为脏数据是最新的）

  Note

  The snoop response does not indicate if a cache is holding a dirty copy of the cache line, it only indicates if the responsibility for updating main memory is being passed.

  Snoop响应并不会标识这个CacheLine是否是Dirty的，只会标识这个CacheLine是否需要被更新到main memory中

- Data read from main memory must not be used if it is possible that the data read is different to the data that would be read after all associated snoop transactions have completed. For example, if a WriteBack or WriteClean transaction to the cache line did not complete before the read from main memory was issued, then the data obtained from main memory must not be used. A new read from main memory must be issued to obtain the correct data.

  WriteBack和WriteClean请求执行结束之前如果有一个read main memory请求被发送执行，那么得到的数据不能直接使用，因为不是最新的，最新的数据还未写入main memory，此时需要等待再次发送一个read main memory的请求才能确保获得了最新的数据

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C6.5.2 Main memory update generated by the interconnect

The following transactions are passed through the interconnect to the appropriate main memory or peripheral device: 这几种请求会直接穿透Interconnect直达main memory或者外设设备（通常是给DMA或者PTW等模块使用的）

- WriteNoSnoop
- WriteBack
- WriteClean.

For the WriteUnique and WriteLineUnique transactions, the interconnect must carry out the required snoop transactions as described in *Issuing snoop transactions* on page C6-229. If a snoop response is received with PassDirty asserted, then either: 如果Write请求并且需要出发Snoop，而Snoop响应中PassDirty拉高，那么此时的数据写入顺序应该是：
（1）首先应该先写入通过Snoop得到的Dirty数据
（2）其次是通过Write请求需要写入的数据

- The order in which data is written must be:
  1. The dirty cache line obtained from the snoop is written to main memory.
  2. The write data that is part of the WriteUnique or WriteLineUnique transaction is written to main memory.

- The write data that is part of the WriteUnique or WriteLineUnique transaction must be merged with the data obtained from the dirty cache line. The valid bytes of the WriteUnique or WriteLineUnique transaction must overwrite the associated bytes of the dirty cache line. A single write to main memory is then performed of the merged data. Write请求需要写入的数据可能需要进行合成（PassDirty的情况下Snoop响应提供了Dirty的数据）因此最终需要写入的数据其实是合成之后的数据（新数据于旧的脏数据的合成）

When a snoop response has the PassDirty response asserted, and the interconnect does not assert the PassDirty transaction response for the initiating master, the interconnect must generate a write transaction to update main memory. This occurs when: 如果Snoop响应中的PassDirty是拉高的，并且Interconnect没有将传给master 设备的PassDirty拉高，那么Interconnect需要产生一个write请求用于更新main memory（脏数据最终都会被写回）

- The transaction from the initiating master does not permit the assertion of the PassDirty response bit. This is true for the following transactions:
  - ReadOnce
  - ReadClean
  - CleanUnique
  - CleanShared.
  - CleanInvalid
  - ReadNotSharedDirty, if the IsShared response is asserted.

- The interconnect has provided a read response to an initiating master before it has received all the snoop responses and a later snoop response has PassDirty asserted.

The interconnect is permitted to carry out a write transaction to update main memory when it receives a snoop response with the PassDirty response asserted. In this case, it must not assert the PassDirty transaction response for the initiating master. Interconnect接收到Snoop响应中如果PassDirty拉高，那么是可以发起一个write 请求来更新main memory的,这种情况下就不需要在发送给master设备的响应中附加PassDirty=1了

Note

- The interconnect must not carry out a write to update main memory if it does not receive a snoop response with the PassDirty response asserted. 如果Snoop响应没有接收到PassDirty，那么是不运行执行write请求来更新main memory的
- This specification recommends that the interconnect does not carry out a write transaction to update main memory unless this is required by the combination of the initiating master transaction type and the received snoop response.

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C6.5.3 Permission to update main memory

The interconnect must ensure that all updates to main memory, both from cached masters and the interconnect itself, are performed in the correct order. The interconnect must only give a cached master permission to update main memory when it is guaranteed that any earlier updates to main memory are ordered.

Permission to update main memory is given to a master by either:

- Giving a transaction response to the master with the PassDirty response asserted.
- Giving a transaction response to the master with the IsShared response deasserted. This gives the master permission to store to the cache line and therefore permission to carry out a write to update main memory.

When a master is given permission to update main memory, the first point at which the master can start the associated write transaction is the cycle after the **RVALID**/**RREADY** handshake in which **RLAST** is asserted for the transaction that gave permission to update main memory. RLAST有效之后才能进行main memory的更新

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